Fredrik Pettit

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Education

University of British Columbia — Vancouver, BC
B.Sc. Computer Science, Artificial Intelligence Option | GPA: 3.8 — *Expected Graduation: May 2028*

Experience

AI Engineer Intern — May 2026 – Sep 2026
AEM — *Hayward, CA*

- Saved **$16K** a year with an AI support agent that escalates unresolved tickets to IT via an idempotent webhook
- Deployed the support agent on Azure Kubernetes with rolling updates, health probes and graceful shutdown
- Raised auto-resolution from 32% to **43%** with a RAG pipeline on Azure using hybrid search over 4,000 pages of company docs, after tracing most escalations to German and acronym-heavy customer questions
- Caught a 6-point drop in correct answers from a silent model update with an automated 300-case regression eval
- Reduced monthly budget reporting from **6+ hours to 5 minutes** with a LangChain agent that turns NetSuite exports into Excel reports via Microsoft Graph API and sends unsure expenses to finance for human review

Software Engineering Intern — Jan 2026 – Apr 2026
ELLUSTROS — *The Blue Mountains, ON*

- Cut failed photo uploads on slow mobile networks by **70%** with presigned URLs that send images straight to S3
- Built the chair recommendation system and backend features for a React Native app serving 300+ users
- Shipped backward-compatible PostgreSQL migrations with Alembic so users on older app versions never broke
- Added Pydantic request tests to GitHub Actions so any breaking API change fails the build before it ships

Teaching Assistant | CPSC 310, CPSC 110 — Aug 2025 – Present
UBC Department of Computer Science — *Vancouver, BC*

- Earned a 4.9/5 average student rating while teaching 900+ students in UBC's largest computer science course

Projects

Email Synced Cashflow Tracker | *Java, Spring Boot, Go, React, PostgreSQL, Redis, Pub/Sub, Docker*

- Built a Spring Boot and React app that tracks orders and profit for a personal resale business with $6K+ in sales
- Turned sale emails into orders with a concurrent Go worker whose writes are idempotent, keyed on order ID
- Sped up the sales dashboard API **36×** by caching report queries in Redis and clearing them on each new order

YouTube Shorts AI Optimizer | *Python, FastAPI, Claude API, Gemini API, MCP, OpenTelemetry, Docker*

- Grew a personal channel to **45M** views with an AI agent that simulates audience attention span and comments
- Shipped an MCP server for **100+** users and cut its p95 latency from 95s to 40s by parallelizing tool calls
- Ran each review exactly once by keying Cloud Run jobs on the video hash so Cloud Tasks retries never repeat work

Apex FSD - StormHacks 2026 (Winner) | *Python, PyTorch, OpenCV, PySDL2*

- Won Best Python Hack at Western Canada's largest hackathon with a fully autonomous driver that plays Forza Horizon by turning a physical wheel in a 100 Hz loop, and hands control back the instant a human grabs it

Scheming - HackMIT 2026 | *Python, Claude API, OpenAI API, Modal, MongoDB*

- Built a lie detector for AI agents playing Werewolf that flags the exact line where a player lies
- Matched human labels on 94% of 300 turns by checking each model's words against its private scratchpad

Technical Skills

Languages & Databases: Java, Python, Go, TypeScript/JavaScript, C++, SQL, PostgreSQL, Redis, MongoDB
Frameworks: Spring Boot, FastAPI, Node.js, React Native, SQLAlchemy, Pydantic, pandas, NumPy, pytest, Jest
Cloud & DevOps: AWS (S3), GCP (Cloud Run, Pub/Sub), Azure, Kubernetes, Docker, Linux, GitHub Actions
AI/ML: PyTorch, TensorFlow, scikit-learn, OpenCV, LangChain, LangGraph, MCP, OpenTelemetry, guardrails
Concepts: Data Structures & Algorithms, Object-Oriented Design, Distributed Systems, Back-End Systems, RESTful APIs, SQL & NoSQL Databases, Performance Optimization, CI/CD, Machine Learning, Agentic AI, Model Evaluation